

figured out how to fund the acquisition of these bikes for themselves. “All of these bikes are invested by HNIs who like to invest in these assets. It is like an alternate asset/investment earning plan similar to how you invest in an FD or a mutual fund or real estate.” Zypp offers 20% pre-tax return to these investors for a 3-year plan, after which it offers to buy them back, essentially turning itself into an EV fintech firm!

In terms of scaling up, Zypp’s

ambitions are simple. Expansion. Be it expanding the number of bikes they currently deploy (which is pegged to reach 15,000 within 12-18 months), or increasing the number of clean deliveries happening in the country, or expanding their operations from Delhi-NCR and Hyderabad to Bengaluru and Mumbai, Gupta aims to do it all.

This is not easy though; the biggest hurdle in Zypp’s plans is the low number of bikes being supplied at present. “The EV ecosystem and

OEMs are still in the nascent stage, so none of the OEMs are able to deliver 1000-2000 bikes a month today, even though we are ready to buy those many bikes every month.”

Additionally, Gupta wants more robust bikes to be made that are suitable for the Indian terrain. Zypp’s outlook is positive, and it believes that with the right nudge, it can propel the country into meeting the clean energy goals it has set.

—Siddha Dhar

3 **NATURAL BATTERIES:** SAFE AND EFFICIENT

While recent incidents of electric vehicle fires have alerted us on battery safety needs, a Jaipur based startup, Natural Battery Tech, is building batteries that it claims can sustain harsh temperatures up to 200°C and still provide optimal efficiency

With the rise in demand for larger battery packs with more safety features, Puneet Jain, CEO, Natural Battery Tech, says they are now designing lithium-ion batteries that can work beyond their rated capacity and also sustain harsh temperatures.

“The lithium cells (NMCs) may get heated and touch up to 55-60 degrees Celsius, leading to thermal runaway. The usual batteries available in the market may use a thermal foam or epoxy sheet that can protect up to 100 or 125 degrees Celsius. By using specialised thermal pads and heat protection devices, we are able to secure the complete pack up to 200 degree Celsius. These materials can also absorb vehicle shocks to safeguard battery internal components,” he explains.

For ensuring safety from faulty cells, the firm has introduced fused connection technology, which isolates the faulty cell in the pack from the rest. “A battery pack is a combination of cells. If you are using a



simple connection type and if any cell goes bad, it cannot isolate itself from the regular pack. While bigger

firms like Tesla go for expensive technologies like ultrasonic bonding to address such issues, we have